

Program 11: Deploying Applications to Azure App Services, Managing Secrets and Configuration with Azure Key Vault, Hands-On: Continuous Deployment with Azure Pipelines.

Step 1: Ensure all recommended settings are enabled under Organization Settings to enforce secure and modern pipeline practices.

The screenshot shows the Azure DevOps Organization Settings page for 'krishnagudi'. The left sidebar contains a search bar and a list of settings categories: General, Security, Boards, Pipelines, and Agent pools. The main content area displays a list of settings, each with a toggle switch and a description:

- Disable anonymous access to badges** (On): Anonymous users can access the status badge API for all pipelines unless this option is enabled (does not apply to public projects).
- Limit variables that can be set at queue time** (On): You can set any variables at queue time unless this option is enabled. With this option enabled, only those variables that are explicitly marked as "Settable at queue time" can be set. [Learn more](#)
- Limit job authorization scope to current project for non-release pipelines** (On): Non-Release Pipelines can run with collection scoped access tokens unless this option is enabled. With this option enabled, you can reduce the scope of access for all non-release pipelines to the current project.
- Limit job authorization scope to current project for release pipelines** (On): Release pipelines can run with collection scoped access tokens unless this option is enabled. With this option enabled, you can reduce the scope of access for all release pipelines to the current project.
- Protect access to repositories in YAML pipelines** (On): Apply checks and approvals when accessing repositories from YAML pipelines. Also, generate a job access token that is scoped to repositories that are explicitly referenced in the YAML pipeline.
- Disable stage chooser** (Off): With this enabled, users will not be able to select stages to skip from the Queue Pipeline panel.
- Disable creation of classic build pipelines** (On): No classic build pipelines can be created / imported. Existing ones will continue to work.
- Disable creation of classic release pipelines** (Off): No classic release pipelines, task groups, and deployment groups can be created / imported. Existing ones will continue to work.

Step 2: After build process, the run failed as no parallelism is granted—students must request free parallelism using the given form.

The screenshot shows the Azure DevOps Pipelines run details page for a failed build. The top navigation bar shows the path: Azure DevOps / krishnagudi / mavenplus / Pipelines / ksgudi.FirstProject / 20250520.1. The left sidebar shows the project structure with 'Pipelines' selected. The main content area displays the run details for '#20250520.1 • Update azure-pipelines.yml for Azure Pipelines'.

Summary

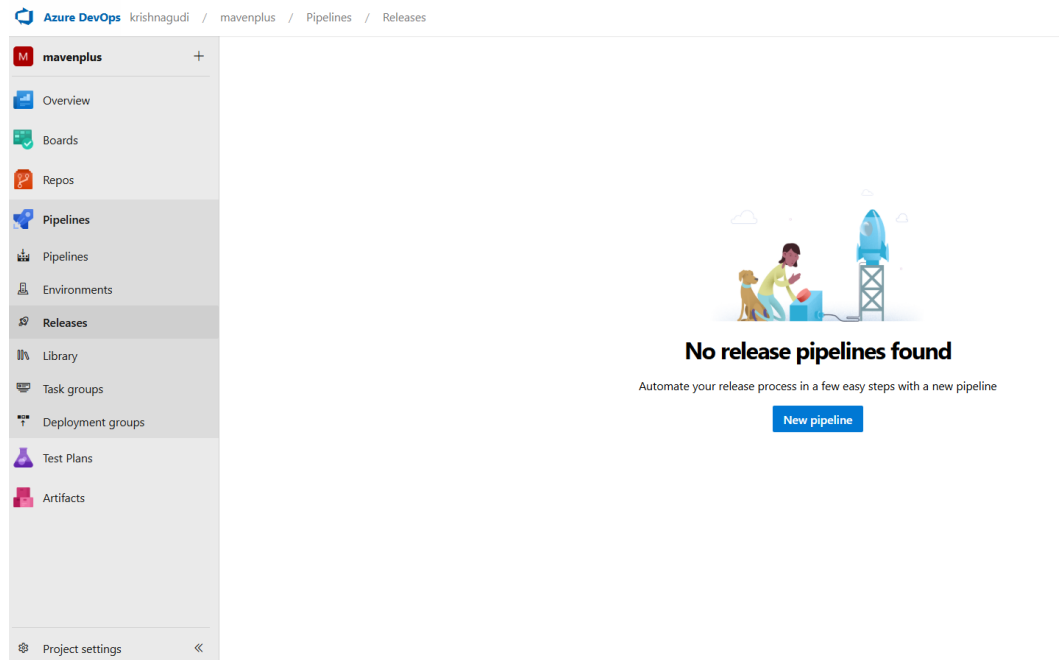
- Manually run by:** krishna gudi
- Repository and version:** ksgudi/FirstProject, master, ced39ab0
- Time started and elapsed:** Just now, <1s
- Related:** 0 work items, 0 artifacts
- Tests and coverage:** [Get started](#)

Errors (1)

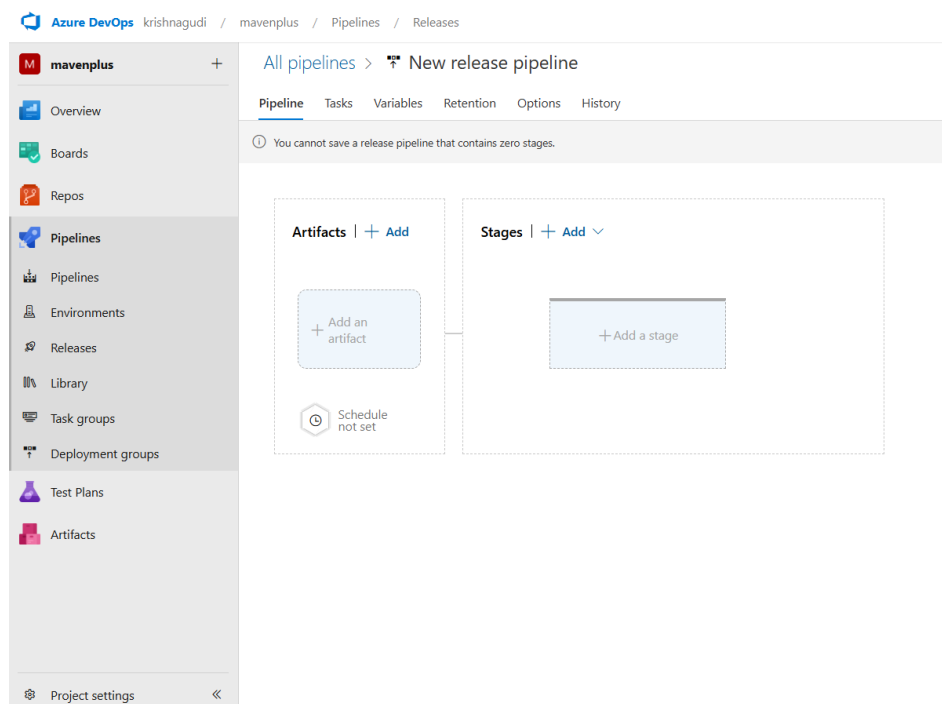
- No hosted parallelism has been purchased or granted. To request a free parallelism grant, please fill out the following form <https://aka.ms/azpipelines-parallelism-request>

[View documentation for troubleshooting failed runs](#)

Step 3: No release pipeline exists—click on ‘New pipeline’ to start creating one for deployment.



Step 4: Begin your release pipeline by adding an artifact and at least one stage to proceed.



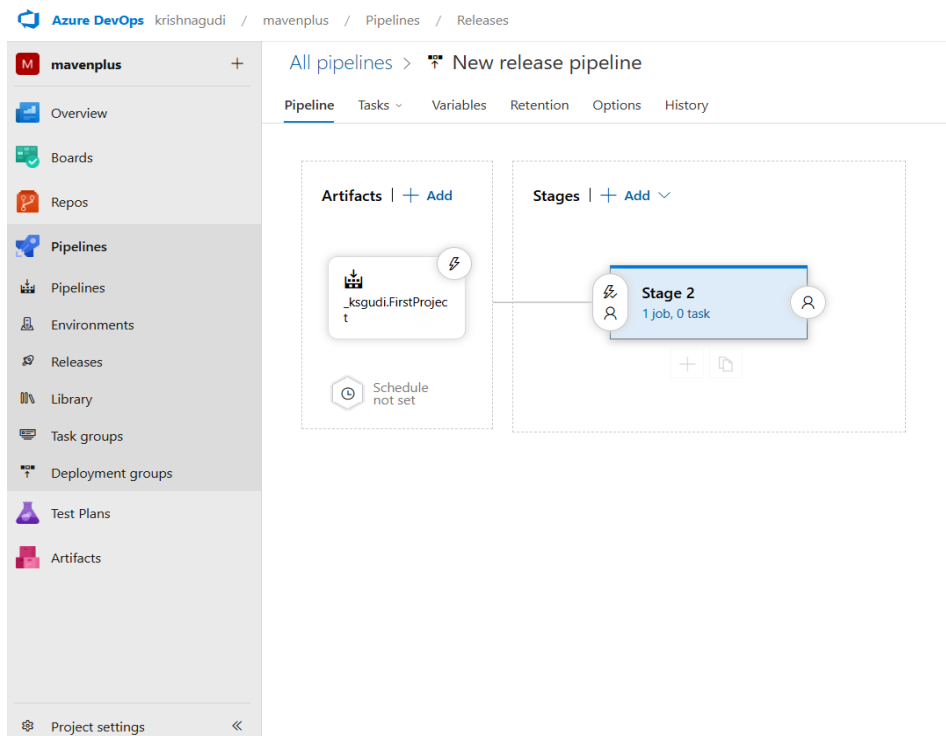
Step 5: Select the build pipeline as the artifact source to link your build output to the release pipeline.

The screenshot shows the Azure DevOps interface for a new release pipeline. The left sidebar contains navigation links for Overview, Boards, Repos, Pipelines, Environments, Releases, Library, Task groups, Deployment groups, Test Plans, and Artifacts. The main area is titled 'New release pipeline' and includes tabs for Pipeline, Tasks, Variables, Retention, Options, and History. A message states: 'You cannot save a release pipeline that contains zero stages.' The 'Artifacts' section shows an 'Add an artifact' button, and the 'Stages' section shows an 'Add a stage' button. On the right, the 'Add an artifact' dialog is open, showing the 'Source type' as 'Build'. The 'Project' is set to 'mavenplus', the 'Source (build pipeline)' is 'ksgudi.FirstProject', and the 'Default version' is 'Latest'. The 'Source alias' is '_ksgudi.FirstProject'. A warning message at the bottom states: 'No version is available for ksgudi.FirstProject or the latest version has no artifacts to publish. Please check the source pipeline.' The 'Add' button is at the bottom right of the dialog.

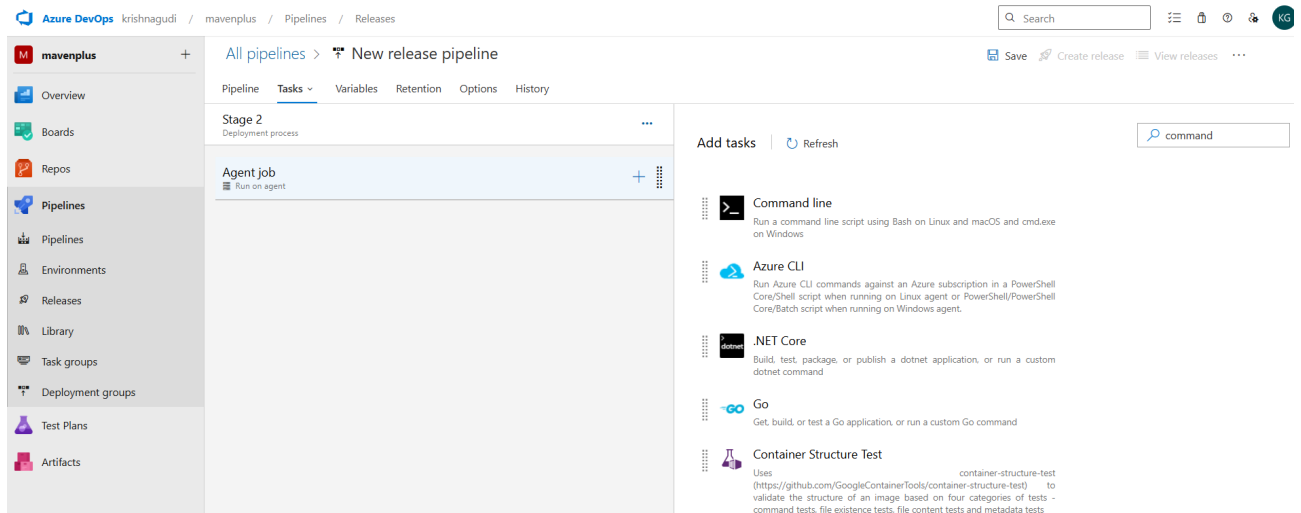
Step 6: After adding the artifact, add a stage and choose a deployment template or start with an empty job.

The screenshot shows the Azure DevOps interface for a new release pipeline. The left sidebar is the same as in Step 5. The main area shows the 'New release pipeline' page with the 'Artifacts' section now containing an artifact named '_ksgudi.FirstProject'. The 'Stages' section shows a 'Stage 2' with a 'Select a template' button. On the right, the 'Select a template' dialog is open, showing a search bar and a list of featured templates. The 'Featured' section includes: 'Azure App Service deployment', 'Deploy a Java app to Azure App Service', 'Deploy a Node.js app to Azure App Service', 'Deploy a PHP app to Azure App Service and Azure Database for MySQL', 'Deploy a Python app to Azure App Service and Azure database for MySQL', 'Deploy to a Kubernetes cluster', and 'IIS website and SQL database deployment'. The 'Others' section is also visible at the bottom.

Step 7: Stage is created—click to configure jobs and tasks for deployment.



Step 8: Under ‘Tasks’, add a new deployment task such as a command line script to simulate or execute deployment steps.



Step 9: In the command line task, enter deployment commands to simulate deployment or perform actual actions.

The screenshot shows the Azure DevOps web interface. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines, Environments, Releases, Library, Task groups, Deployment groups, Test Plans, and Artifacts. The main area is titled 'New release pipeline' and shows 'Stage 2' with a 'Deployment process' tab. Under 'Agent job', a 'Command Line Script' task is selected. The right-hand configuration panel for this task shows 'Task version' set to '2.*'. The 'Display name' is 'Command Line Script'. The 'Script' field contains the text: 'echo "Simulated Deployment Step"' and 'echo "Artifact would be deployed here"'. Below the script field are expandable sections for 'Advanced', 'Control Options', 'Environment Variables', and 'Output Variables'.

Simulate Secrets (Key Vault):

Step 10: Click on '+ Variable group' to create a new group of shared pipeline variables.

The screenshot shows the 'Library' page in Azure DevOps. The left sidebar is the same as in the previous image. The main area has tabs for 'Variable groups', 'Secure files', '+ Variable group', 'Security', and 'Help'. The 'Variable groups' tab is active. In the center of the page is a large graphic with a box containing '(x)'. Below this graphic, the text reads 'New variable group' and 'Create groups of variables that you can share across multiple pipelines.' There is a blue button labeled '+ Variable group' and a link that says 'Learn more about variable groups.'.

Step 11: Enter a name and define secure variables (e.g., DB_PASSWORD) for use across multiple pipelines.

Azure DevOps krishnagudi / mavenplus / Pipelines / Library

mavenplus +

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Releases

Library

Task groups

Deployment groups

Test Plans

Artifacts

Project settings <<

Library > New variable group 20-May*

Variable group | Save | Clone | Security | Pipeline permissions | Approvals and checks | Help

Properties

Variable group name

New variable group 20-May

Description

☐ Link secrets from an Azure key vault as variables ⓘ

Variables

Name ↑	Value	
name (required)		

+ Add

Step 12: Fill in variable name and value, and click Save to finalize the variable group setup.

Azure DevOps krishnagudi / mavenplus / Pipelines / Library

mavenplus +

Overview

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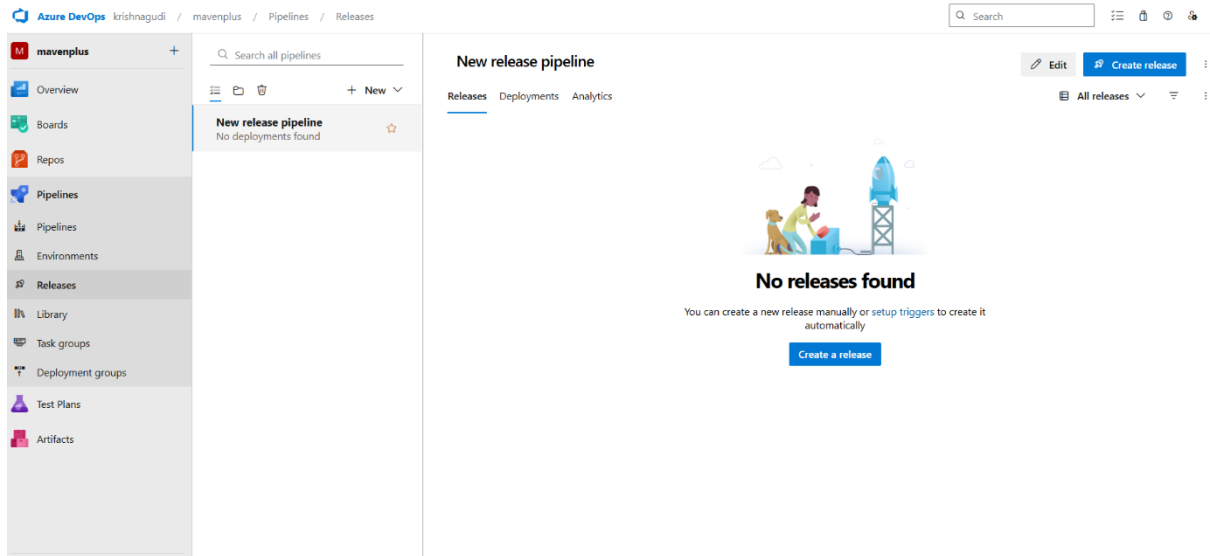
Variables

Name ↑	Value	
DB_PASSWORD	xyz123	

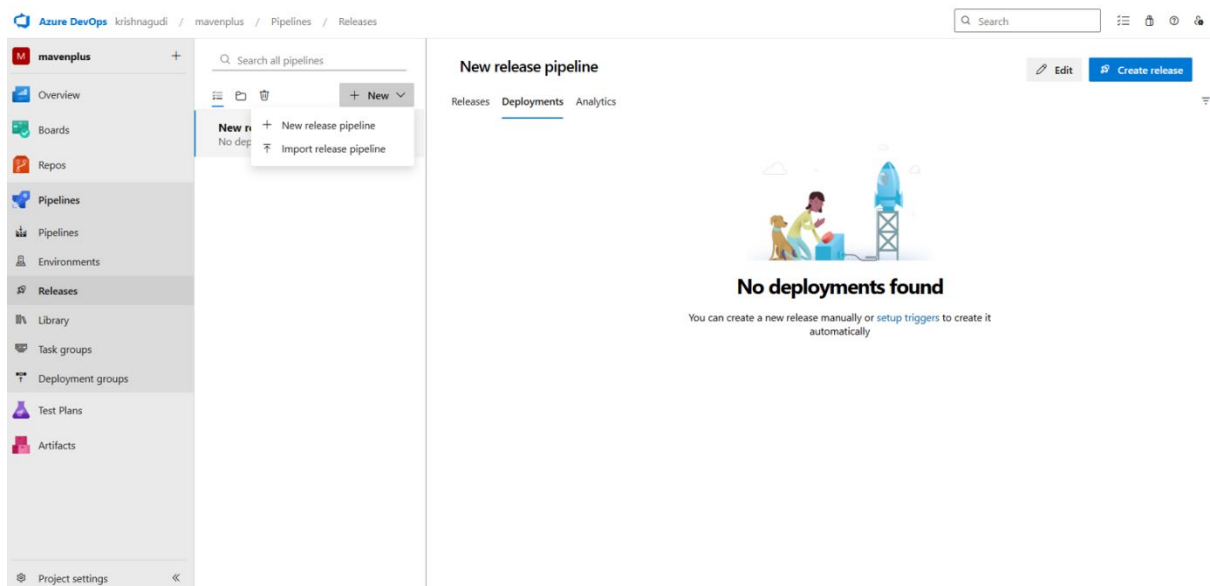
+ Add

Link this variable group in your Release Pipeline:

Step 13: Select the created variable group and click ‘Link’ to use it in the release pipeline.



Step 14: Click on ‘Create a release’ to trigger the release pipeline manually.



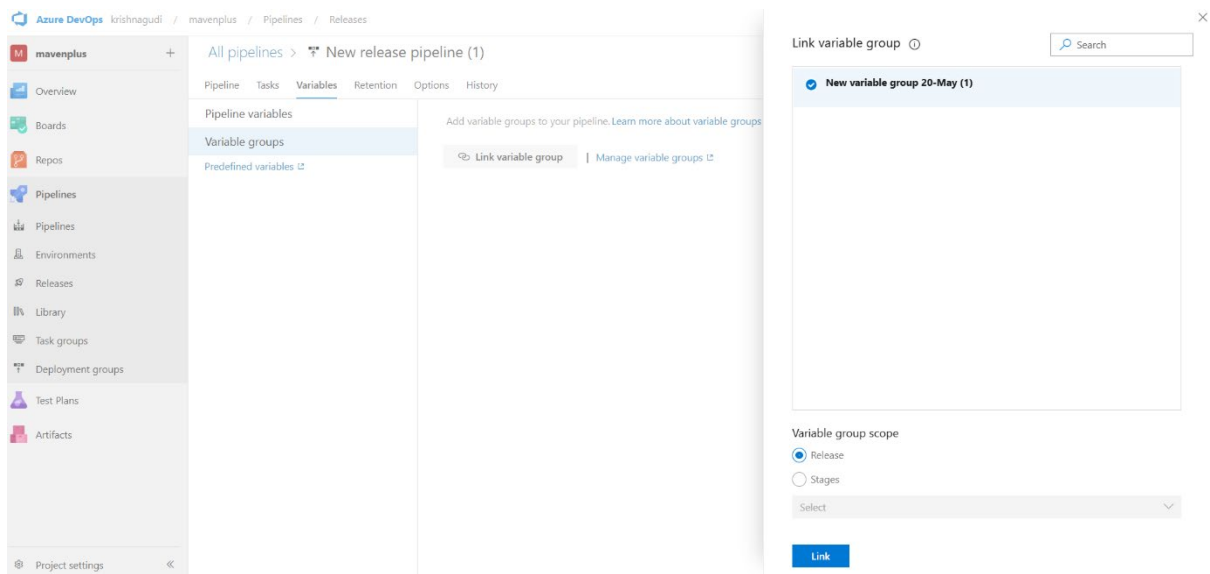
Step 15: Click the ‘+ New’ button to create or import a release pipeline when none exists.

The screenshot shows the Azure DevOps interface for a project named 'mavenplus'. The left sidebar contains a navigation menu with icons for Overview, Boards, Repos, Pipelines, Environments, Releases, Library, Task groups, Deployment groups, Test Plans, and Artifacts. The 'Pipelines' section is expanded, showing a '+' button. The main area displays the 'New release pipeline (1)' page. The 'Variables' tab is selected, showing 'Pipeline variables' and 'Variable groups'. A '+ Add' button is visible under 'Pipeline variables'.

Step 16: Click ‘+ Add’ under Pipeline variables to define new custom key-value variables.

The screenshot shows the Azure DevOps interface for a project named 'mavenplus'. The left sidebar contains a navigation menu with icons for Overview, Boards, Repos, Pipelines, Environments, Releases, Library, Task groups, Deployment groups, Test Plans, and Artifacts. The 'Pipelines' section is expanded, showing a '+' button. The main area displays the 'New release pipeline (1)' page. The 'Variables' tab is selected, showing 'Pipeline variables' and 'Variable groups'. A '+ Add' button is visible under 'Pipeline variables'.

Step 17: Use ‘Link variable group’ to connect predefined variables to your release pipeline.



“To overcome subscription-related limitations, a simulated deployment approach was adopted using command line scripts. This enabled demonstration of the complete release pipeline flow without requiring actual agent execution. It served as an effective workaround in a restricted environment.”